

NETWORKS AND COMPLEXITY

Solution 12-9

*This is an example solution from the forthcoming book *Networks and Complexity*.*

Find more exercises at <https://github.com/NC-Book/NCB>

Ex 12.9: Price's model [Lvl. 3]

When a scientific paper is published it cites m other papers. Papers that have already been cited more often have higher visibility and hence attract new citations at a higher rate. We assume that a paper with k citations has the visibility $v = a + k$, where a represents an intrinsic visibility independent of citations.

- a) Let p_k be the in-degree distribution of the papers, the proportion of papers that have been cited k times. Consider a single citation from a new paper. Assume that the chance that a paper is cited is proportional to its visibility. Show that the probability v_k that a single new citation cites a paper that has previously had k citations is

$$v_k = p_k \frac{a + k}{a + z}$$

Solution

From the question we know that the probability is proportional to the visibility $a + k$, it should also be proportional to the proportion of target nodes with the respective degree. So we can use the ansatz

$$v_k = A p_k (a + k) \quad (1)$$

where A is a factor that we have to pick such that the distribution is normalized, i.e.

$$1 = \sum v_k = \sum A p_k (a + k) = A(a + z) \quad (2)$$

where z is the mean in-degree. We can now read off

$$A = \frac{1}{a + z} \quad (3)$$

This means

$$v_k = p_k \frac{a + k}{a + z} \quad (4)$$

- b) Show that in the long run the in-degree distribution p_k approaches a power-law for $k \gg 1$, and determine the power-law exponent.

Solution

Following the same reasoning as for the BA-model in the lecture yields the ODE

$$\dot{c}_k = \frac{1}{N} \left(-c_k + \frac{m p_k (a + k)}{a + z} \right) \quad (5)$$

Here we need to be careful because the mean in-degree will only approach $z = m$ instead of $z = 2m$ for the undirected network. Therefore,

$$\dot{c}_k = \frac{1}{N} \left(-c_k + \frac{mp_k(a+k)}{a+m} \right) \quad (6)$$

In the steady state

$$c_k = \frac{mp_k(a+k)}{a+m}. \quad (7)$$

and hence

$$p_k = c_k \frac{a+m}{m(a+k)} \quad (8)$$

which together with $\partial_k c_k = -p_k$ yields

$$\partial_k c_k = -c_k \frac{a+m}{m(a+k)}. \quad (9)$$

We integrate

$$\int \frac{1}{c_k} dc_k = -\frac{a+m}{m} \int \frac{1}{a+k} dk \quad (10)$$

$$\ln(c_k) = -\frac{a+m}{m} \ln(a+k) + C \quad (11)$$

$$c_k = A(a+k)^{-1-a/m} \quad (12)$$

where $A = e^C$. Hence we can compute

$$p_k = -\partial_k c_k = B(a+k)^{-2-a/m} \quad (13)$$

where $B = A(1+a/m)$ is another prefactor. In the limit $k \gg a$ we get the power law

$$p_k \sim k^{-2-a/m} \quad (14)$$

So, the distribution of citations that we get from Price's model follows a power law with exponent $\gamma = -2 - a/m$.